

Cover Letter

Date: 2026-09-23 To: Editor-in-Chief, Advances in Space Research From: LIU Shuhao, College of Intelligent Systems Science and Engineering, Harbin Engineering University, liushuhao@hrbeu.edu.cn

Dear Editor,

I am submitting the manuscript “Physics-Aware Multi-Objective Scheduling for an Agile SAR Satellite: Balancing Profit, NESZ, and Geometric Resolution” for consideration as a full-length article in Advances in Space Research.

This manuscript falls within the journal’s scope of space technology applications for Earth observation: it studies the scheduling of an agile SAR satellite whose imaging quality is coupled to observation geometry, a problem at the intersection of spacecraft resource management and SAR remote sensing instrumentation.

The manuscript presents a physics-aware multi-objective optimization framework for single-satellite agile SAR scheduling. Five configurations spanning greedy heuristics, a single-objective GA control, and NSGA-III variants are compared on 200 systematically varied scenarios across four density classes ($N = 20$ to 500). The central finding is a conditional empirical saturation regime for SAR quality optimization: G-BL-seeded NSGA-III converges to low-squint geometries under which the three-objective formulation is indistinguishable from the two-objective one at operational densities, while retaining measurable quality gains at sparse density. The work establishes that multi-objective scheduling provides measurable quality benefit only when observation requests are sparse ($N \lesssim 100$), within the single-satellite, single-pass Stripmap setting studied. A matched-knee-point control confirms the sparse-density third-objective gain is a modest controlled margin (+4.9% under matched knee selection), not an artifact of NSGA-III reference-direction count or knee selection.

Disclosure of related manuscript: I have an article previously accepted by Advances in Space Research entitled “Event-Triggered Attitude Decision-Making for Agile SAR Satellite Formations: A Self-Triggered Replanning Approach” (Ch4 of the author’s doctoral thesis). That work addresses event-triggered attitude control decisions for SAR satellite formations, which is an orthogonal topic to the present manuscript. The two works share no overlapping experimental results, solver implementations, or research claims. The current manuscript focuses on single-satellite scheduling with imaging quality objectives; the accepted work addresses formation-level attitude replanning triggered by observation events. I confirm that the present manuscript contains no text or figures from the accepted article, and that the two manuscripts are distinct contributions to the author’s doctoral thesis. Elsevier’s same-journal policy permits multiple articles from the same thesis when they address different aspects of the research.

This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have approved the submission.

Thank you for your consideration.

Sincerely,

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